



GE Medical Systems

Technical Publications

Direction 2100619

Revision 1

Signa[®] Advantage[™] 1.0T Auto Tune Device II

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Operating Documentation

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation "damage in shipment" written on all copies of the freight or express bill before delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, nature, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

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<u>REVISION</u>	<u>DATE</u>	<u>PRIMARY REASON FOR CHANGE</u>
REV 0	1/28/94	Initial Release
REV 1	6/17/96	Add 60 cm values to Table 2-2

<u>PAGE</u>	<u>REV</u>	<u>PAGE</u>	<u>REV</u>	<u>PAGE</u>	<u>REV</u>	<u>PAGE</u>	<u>REV</u>
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SECTION 1 - INTRODUCTION

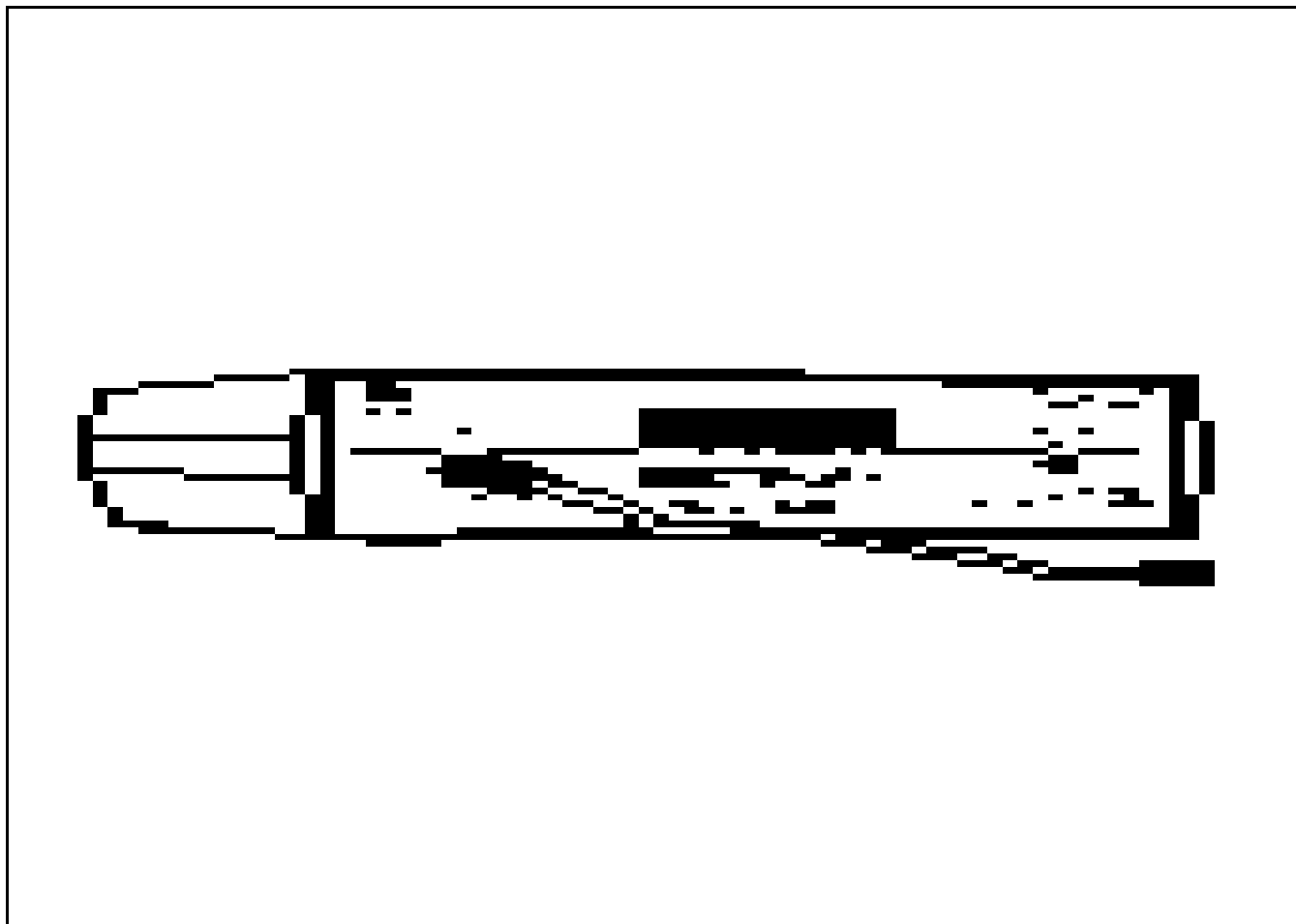
1-1 THE ATD-II

The ATD-II is to be used with the Signa 1.0T system. Connection to the scanner Head Coil Port will be with an integral cable. See ILLUSTRATION 1-1.

1. Overview

The system consists of an Auto-Tune Device II [ATD-II] which provides the interface, tuning, decoupling, and coil checking functions required to operate disposable coils with the GE Signa Advantage™ 1.0 Tesla MRI scanner system.

The ATD-II electronically optimizes the tuning of the disposable coil on a per-coil and per-patient basis by sweeping an electronically controlled reactance [varactor diode] through a range of values, and locking in the value which provides the optimum response at the nominal 42.68 MHz MRI system frequency. This allows the use of an inexpensive, disposable coil which does not include internal tuning components. The components required to provide effective decoupling of the disposable coil from the RF excitation field during transmit are included in the ATD-II.



ATD-II
ILLUSTRATION 1-1

1-2 OPERATION DESCRIPTION OF THE ATD-II

See ILLUSTRATION 1-2

1. Coil Tuning/Decoupling Network

The network consists of a parallel tuned circuit which includes a varactor diode. Tuning is achieved by automatically setting the voltage on the varactor diode. At the output of the coil tuning network, the ATD-II includes a low noise FET preamplifier with a gain of about 16 dB.

A pair of crossed high speed switching diodes are connected across the coil input port. During transmit by the MR system, the RF voltage induced in the coil causes the diodes to conduct, placing a very low impedance across the coil output cable. The cable appears as a very high impedance. This provides a blocking impedance in the coil loop to decouple the coil from the incident RF transmit field.

The endorectal coil cable length is about 60 degrees rather than 90 degrees long at the system operating frequency. To create an electrical length of 90 degrees, a 50 ohm network having a delay of 30 degrees is included in series with the endorectal coil input.

2. PIN Diode Switch

A low loss PIN diode switch at the output of the FET preamplifier connects the preamplifier output to either an RF signal level detector during the tuning procedure, or to the cable to the SURFACE COIL INPUT PORT of the Signa MRI System.

3. 42.68 MHz Signal Source

The ATD-II contains a crystal controlled RF source which radiates a signal at the 42.68 MHz near the coil input network. This source is enabled only during the execution of the automatic tuning function.

4. Description of Tuning Cycle

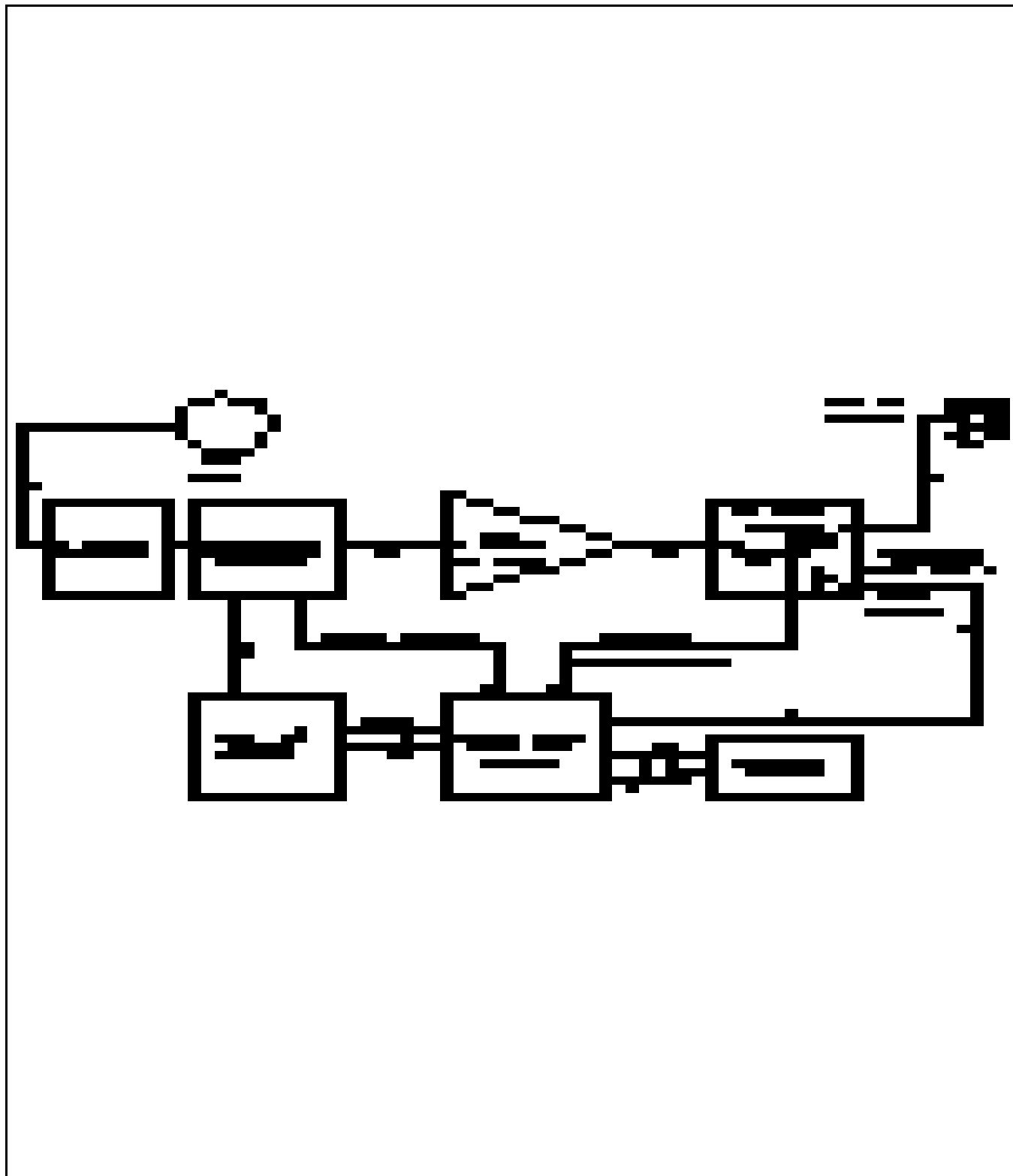
The ATD-II uses a logic and sequence controller to regulate and coordinate the operation of all the internal functions. The manner of operation is as follows:

The ATD-II, upon connection of the disposable coil, begins ramping the tuning through the range until a maximum signal amplitude from the preamplifier is determined. The varactor voltage is locked at that point. A window comparator verifies that the coil is performing properly by checking that the signal amplitude falls within the acceptance window.

In the event that either no maximum is found in the signal response to tuning, or that the signal amplitude falls outside of the acceptance window, the device displays a **FAIL** message to alert the operator that a coil failure has been detected.

If an acceptable signal maximum is found, the ATD-II shuts off the RF signal source, switches the output of the FET preamplifier to the MRI scanner output cable, and displays a checkmark message that the tuning has been successfully completed, and the Endorectal Coil System is ready to operate in the **SCAN** mode.

1-2 OPERATION DESCRIPTION OF THE ATD-II - Continued

ATD-II BLOCK DIAGRAM
ILLUSTRATION 1-2

1-2 OPERATION DESCRIPTION OF THE ATD-II - Continued**5. Battery Condition Detector**

The ATD-II contains a level comparator to determine the state of charge of the internal battery. In the event that the battery is not above 10.5 volts at the beginning of the tuning cycle, the display indicator will provide a message indicating **LOBAT** rather than the usual **TUNE** message. In the event the battery level drops below the threshold during the scan procedure, the comparator does not interrupt the scan, as it is active only at start-up. Expected battery life between charges is about six hours of operation when beginning with a fully charged battery.

6. Readout Display

During a normal coil tuning cycle, the display reads **TUNE** to alert the operator that the **TUNE** function is occurring. The display switches automatically to either the **FAIL** or a checkmark at the end of the **TUNE** function, within 10 seconds of the beginning of the cycle. The display is steady state rather than clocked or multiplexed to ensure that no RF noise is generated during scanning.

7. Device Power

The **OFF** and **ON** functions of the ATD-II are controlled by the presence or absence of the Signa Endorectal Coil at the **PROBE INPUT** port of the device. Inserting the coil connector into the receptacle on the ATD-II activates the device and begins the tuning cycle. Removal of the coil connector will deactivate the ATD-II. At all times when it is operational, the status of the ATD-II is indicated by a message in the status display.

1-3 COMPATIBILITY

The ATD-II is compatible with the following hardware configurations:

- GE SIGNA ADVANTAGE™ 1.0T Systems

1-4 RELATED DOCUMENTS

- Direction 15400 - Signa Advantage 1.5T, 1.0T, and 0.5T Systems
- Direction 15404 - Signa Advantage 1.5T, 1.0T, and 0.5T Renewal Parts.

1-5 ORGANIZATION OF THIS DOCUMENT

This document is divided into the following sections:

Section 1, INTRODUCTION, describes what the ATD-II is and where the ATD-II can be used.

Section 2, SET UP AND CALIBRATION, describes installation procedures

Section 3, FUNCTIONAL CHECKS, describes normal power-up sequence.

Section 4, REPLACEMENT/MAINTENANCE, describes field maintenance procedures.

Section 5, RENEWAL PARTS, lists field replaceable service kits and parts.

SNR Data Table

Coil Return Form

TLT Table

1-6 ENVIRONMENTAL REQUIREMENTS

Operate and store the ATD-II in the Scanner Room. The Battery Charger/Charging Stand should not be brought into the scanning room.

SECTION 2 - SET UP AND CALIBRATION

2-1 CHECKING THE SHIPPING LIST

Table 2-1 lists the M1885EE Signa ATD-II parts. Check that all parts have been shipped.

TABLE 2-1
COIL PARTS

QTY.	ITEM	PART NO.
1	ATD-II	46-328293P1
1	Battery Charger	N/A
1	Endorectal Coil Simulator	2100937-3
1	Service Manual	2100619

2-2 INSTALLING THE ATD-II

At the console, install a new soft key. This key will be used by the operator to select ATD-II imaging. Name the key "Endorec".

Refer to the Signa Advantage 1.5T, 1.0T, & 0.5T System Manual for information on installing soft keys (use Coil default values in Table 2-2).

Set the Battery Charger/Charging Stand to the proper input voltage. See Section 3, FUNCTIONAL CHECKS.

TABLE 2-2
COIL VALUES

SYSTEM	COIL NAME	COIL TYPE	EXTREMITY COIL	CABLE LOSS	COIL LOSS	RECON SCALE FACTOR	XMIT COIL TYPE	XMIT ATTN	MULTI-COIL.
1.0T	Endorec	surface	no	1.24	0.103 - 55cm 0.549 - 60cm	Head Coil Recon Scale Fctr. x 0.25	Quad	0	no

2-3 FUNCTIONAL CHECKS

1. Perform a Body coil scanner performance verification. Refer to Section 3-1, SIGNA ADVANTAGE 1.0T SCANNER VERIFICATION.
2. Perform a ATD-II performance verification. Refer to Section 3-2, ATD-II IMAGING PERFORMANCE VERIFICATION.
3. Perform an SNR Image analysis. Refer to Section 3-3, SNR IMAGE ANALYSIS.

2-4 PERIODIC QUALITY ASSURANCE CHECK

On a periodic basis, such as during planned maintenance, perform the quality assurance check outlined below to ensure that the ATD-II is operating properly with no appreciable degradation of image quality.

1. Check the external cable for cracks or breaks once each week.
2. Perform a ATD-II performance verification. Refer to Section 3-2, ATD-II IMAGING PERFORMANCE VERIFICATION.
3. Record the date and value calculated in SNR QA DATA TABLE.
4. As instructed in the Data Table, divide the SNR value obtained in the periodic QA check by the original SNR value.
5. If this ratio is not greater than 85%, then there may be a problem with the ATD-II or the MR system. Contact the GE Service Representative.

SECTION 3 - FUNCTIONAL CHECKS

3-1 SIGNA ADVANTAGE™ 1.0T SCANNER VERIFICATION

Note

An alternative proprietary procedure is available for GE use and to customers with a valid Advanced Service Package Limited License. Refer to "TLT PROCEDURE" located in Direction15491, *Signa Advantage 1.5T, 1.0T, and 0.5T System Troubleshooting Guide*.

1. Scanner performance in the body coil mode should be verified. The RF power level calibration from the body coil mode is used to verify the surface coil decoupling. Landmark the decoupling phantom without the endorectal coil simulator.
2. Set up the scan using the parameters in TABLE 3-1. Coil Type: **[Body Coil]** and Number of Echos: **[4]** should be inserted. Perform an **Auto Prescan** to calibrate the RF power level. Record the value for TG, R, and R2.
3. Run the scan. Observe the resulting images. Ensure that there are no artifacts of any sort on any of the images.
4. Select **[Cancel]**, to perform SNR comparisons. Acquire two identical images using the previously loaded scan parameters in TABLE 3-1 after inserting Number of Echos: **[1]**. Immediately after acquiring the first image, select **Scan** to obtain a second. Note the series number for future reference.

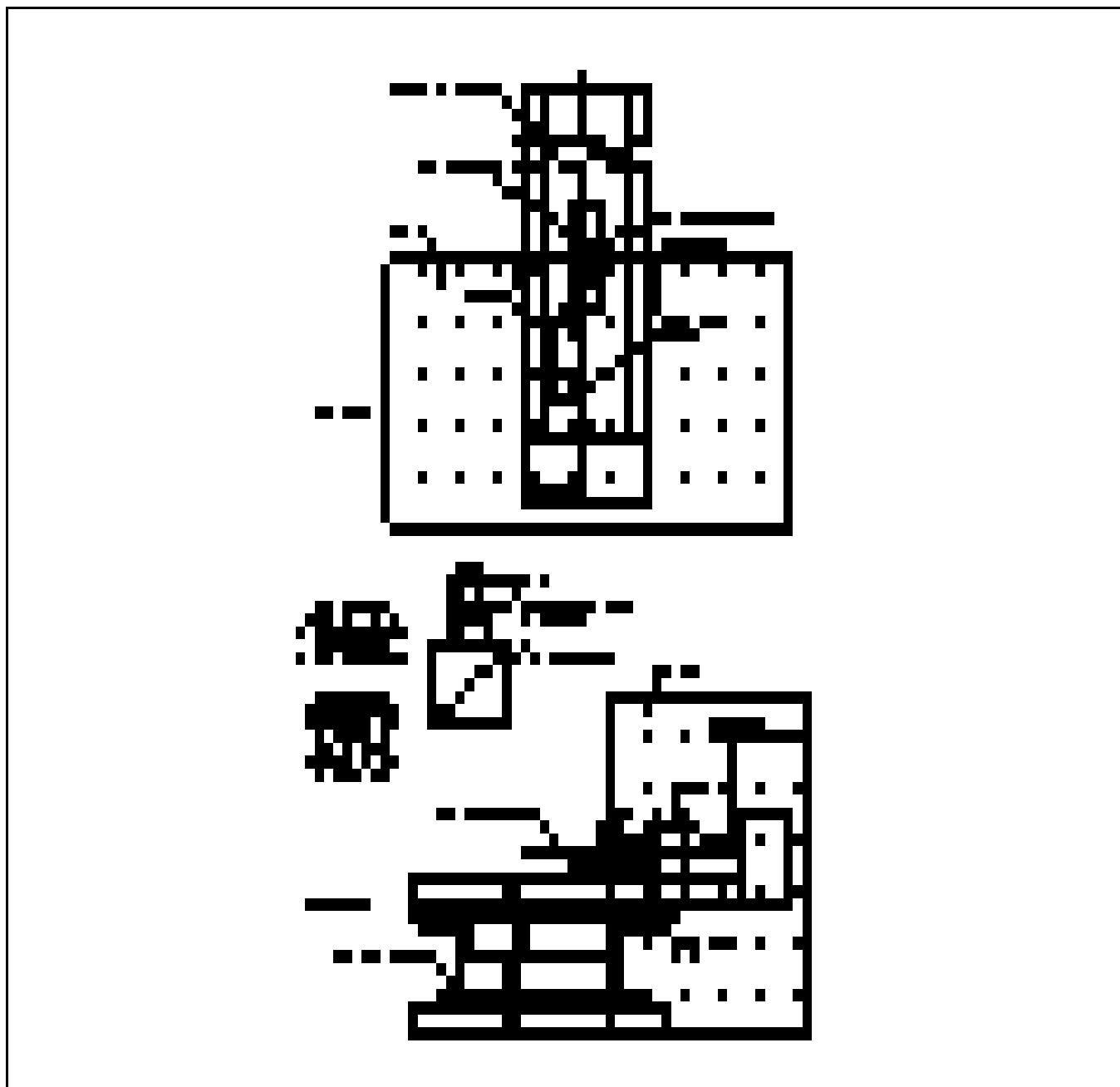
TABLE 3-1
SCAN PARAMETERS

ID:	geservice	Auto CF:	[Peak]
Name:		Receive Bandwidth:	[16kHz]
Weight (lb):	111	Field of View:	[24 cm]
Patient Entry:	[Head First]	Slice Thickness:	[10 mm]
Patient Position:	[Supine]	Interscan Spacing:	[0]
Axial/Sag Landmark:	[Sternal Notch]	Start Location (I/S):	0
Coil Type:	[++] See Note	End Location (I/S):	0
Scan Plane:	[Axial]	No of Scan Locations:	1
Image mode:	[2D]	FOV Center (L/R):	0 (P/A): 0
	[Monitor SAR]	Acq. Matrix: (freq.)	[256]
Pulse Sequence:	[Spin Echo]	Acq. Matrix: (phase)	[128]
Imaging Options:	[None]	Frequency Direction:	[R/L]
Number of Echos:	[*] See Note	Phase FOV:	default
Echo Time (TE):	[20 ms]	Imaging Time:	[1 Nex]
Rep Time (TR):	[Other] 600 ms	Contrast:	[No]
		Table Delta	0

NOTE: * **[4]** for Section 3-1-2; 3-2-2
[1] for Section 3-1-4; 3-2-4
 ++ **[Body]** for Section 3-1-2; 3-2-2
[Other][Endorec] for Section 3-1-4; 3-2-

3-2 ATD-II IMAGING PERFORMANCE VERIFICATION

1. Set up the ATD-II and endorectal coil simulator as shown in ILLUSTRATION 3-1. Connect the ATD-II cable to the **SURFACE COIL QUICK DISCONNECT BOX** and connect to the **HEAD COIL PORT**.
2. Select **[Cancel]**, **[New Series]**, and repeat the scan using the parameters in TABLE 3-1. Coil Type: **[Other][Endorec]** and Number of Echos: **[4]** should be used. Use **Auto Prescan** to set the values of R1 and R2 to those in Section 3. If the value of TG does not match the value obtained in step 3-1-2, use **Manual Prescan** to set the value of TG.

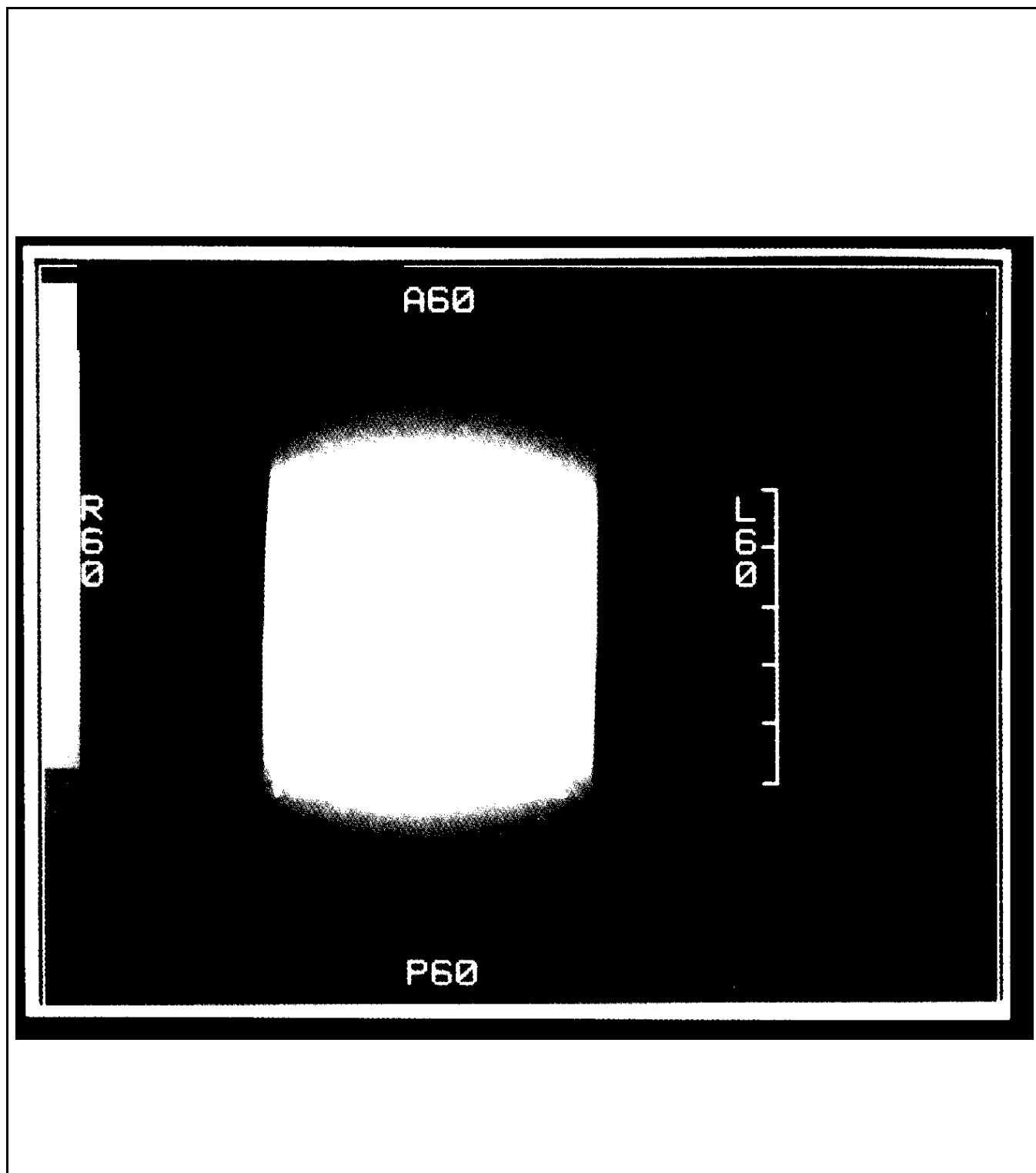


TLT PHANTOM PLACEMENT
ILLUSTRATION 3-1

3-2 ATD-II IMAGING PERFORMANCE VERIFICATION - Continued

3. Run the scan. Ensure that there are no artifacts of any sort on any of the images. Study the image using a **Window Width** and **Window Level** to obtain good contrast. A properly functioning coil will produce an image with a smooth signal pattern, gradually dropping in intensity as the distance from the coil increases, see ILLUSTRATION 3-2. A decoupling failure will cause holes, distortions, or striped patterns in the image, Refer to Section 3-4 to 3-6 to troubleshoot decoupling failures.
4. Repeat step 2 with Number of Echos: **[1]** inserted. Run the scan. Immediately after acquiring the first image, select **Scan** to obtain a second.
5. Compute the SNR using the procedure described in Section 3-3, SNR IMAGE ANALYSIS. Compare the signal to noise ratio of the first echo of the ATD-II images with the signal to noise ratio of the first echo of the body coil images of step 3-1-4. The ATD-II scans should display a signal to noise ratio greater than 2.6 times that of the body coil scans.
6. Inspect the images for visible ghosting (similar in appearance to motion artifacts) in the phase encoding direction. If images contain ghosting artifacts, suspect an intermittent cable or connection. Refer to Section 3-4 to 3-6 for troubleshooting information.

3-2 ATD-II IMAGING PERFORMANCE VERIFICATION - Continued



NORMAL DECOUPLING
ILLUSTRATION 3-2

3-3 SNR IMAGE ANALYSIS

Description

The SNR tool retrieves two operator selected images. Signal value is computed as the mean pixel value in a ROI covering 80% of the image. The image is analyzed to determine the center for positioning the ROI. A difference image is created by subtracting the second image from the first and to calculate noise from the subtracted image. Signal value, noise value, and SNR are reported. The difference image can be saved with the results annotated (option).

Procedure

1. Touch **[Utilities]**, **[MR Tools]**, **[Image Quality]**, then **[SNR Test]**. The SNR Test screen is displayed. See ILLUSTRATION 3-3.
2. Enter first image exam, series, and image numbers. If exam, series, or image numbers are not known, select **[List Exams]**, **[List Series]**, or **[List Images]** to display list to choose from. Enter second image series and image number. The second image Exam Number is defaulted to the same as first image.

Note

Image number selection must be back lit (highlighted) to be able to enter information. Use Switch key on keyboard to transfer control from left to right side of Touch Screen.

3. Select **[Filter Off]** and **[Image Off]**. Touch **[Accept]** to begin analysis. The final values are displayed on the screen.
4. Record Signal to Noise Ratio on SNR QA DATA TABLE. Touch **[Continue]**, select the next exam and repeat for each pair.

NEMA STANDARD SNR TEST				
	Filter Off	Image Off		
First Image	List/Sel Exams	List/Sel Series	List/Sel Images	
Second Image		List/Sel Series	List/Sel Images	
Cancel Utility Main MR Tools Main Image Quality Accept				

Note: Accept changes to continue only after an analysis has been performed.

SNR TEST SCREEN
ILLUSTRATION 3-3

3-4 TUNING TEST

1. The tuning test is normally performed in an open area outside of the scan room. Use the Endorectal Coil Simulator for the following tests.
2. Position the Endorectal Coil Simulator on a nonmetallic support of two comfort pads in a stack. Ensure that no other objects are within 12 inches of the phantom.
3. Insert the connector of the Endorectal Coil Simulator into the **PROBE INPUT** Jack on the ATD-II. Observe the Readout Display. The **TUNE** message should be visible for a period not exceeding ten seconds, followed by a checkmark, indicating that the ATD-II has successfully tuned the coil.

3-5 PIN DIODE TEST

1. Select the DIODE TEST function on the Digital Multimeter.
2. Using a BNC to Banana Plug adaptor, connect the **NEGATIVE** lead of the DMM to the shell of the BNC connector on the cable of the ATD-II. Connect the **POSITIVE** lead to the center pin of the connector on the ATD-II cable.
3. A reading of 0.400 to 0.900 should be observed on the DMM
4. Using a BNC to Banana Plug adaptor, connect the **POSITIVE** lead of the DMM to the shell of the BNC connector on the cable of the ATD-II. Connect the **NEGATIVE** lead to the pin of the BNC connector on the ATD-II cable.
5. A reading of **INFINITY** should be observed on the DMM.
6. If a reading below 0.400 is observed in either direction, either the output cable is shorted or the PIN diode in the ATD-II is shorted.
7. If a reading of above 0.900 is observed in step 3-5-3, the PIN diode is defective.
8. If a reading of **INFINITY** is observed in both directions, either the PIN diode or the cable in the ATD-II is open.

3-6 OUTPUT CABLE TEST

1. Repeat the tests in Section 3-5 while flexing the ATD-II cable, especially near the BNC connector and the strain relief on the front panel of the ATD-II. See ILLUSTRATION 3-4.
2. If the cable fails any of the above tests, it should be replaced. See Section 4-2, EXTERNAL COAXIAL CABLE REPLACEMENT for cable replacement.

3-6 OUTPUT CABLE TEST - Continued



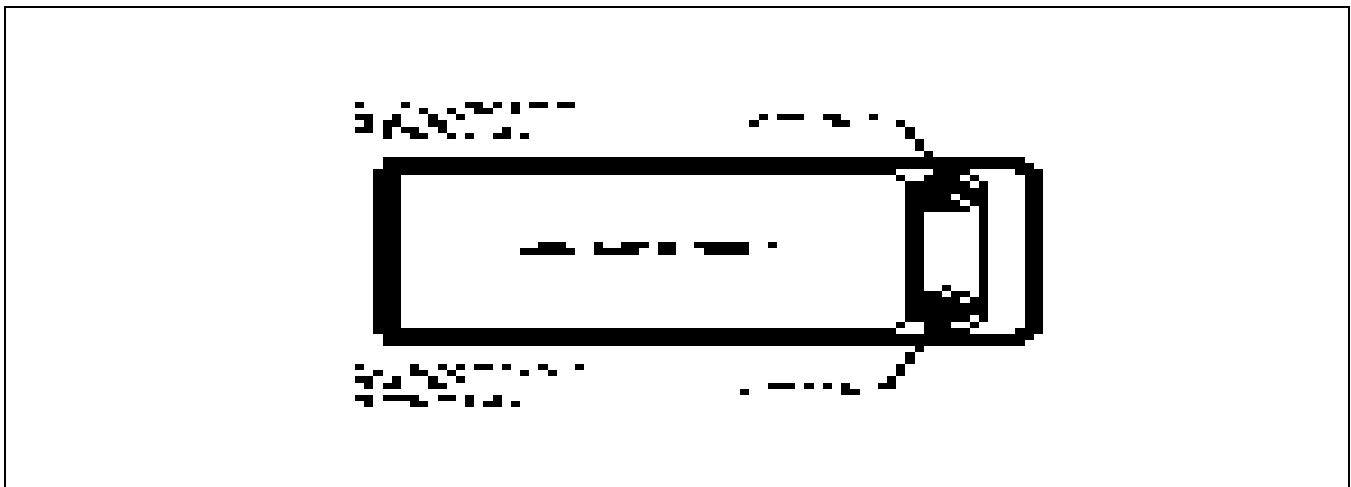
OUTPUT CABLE TEST
ILLUSTRATION 3-4

3-7 ATD-II BATTERY MODULE CONDITION TEST

NOTE

Module contains a lead-acid battery.

1. Ensure that the ATD-II Battery Module is fully charged. Place the ATD-II Battery Module into the Battery Charger supplied with the ATD-II. Insert the Battery Charger into an appropriate AC power source, and allow the ATD-II Battery Module to charge until the charging indicator on the Battery Charger extinguishes, indicating a fully charged condition. This may take up to six hours, depending on the initial charge level of the battery.
2. If a fully charged condition cannot be obtained, see Section 3-8, BATTERY CHARGER TEST to test the Battery charger. If the Battery Charger is performing properly, the ATD-II Battery Module is likely to be defective.
3. Remove the fully charged ATD-II Battery Module from the Battery Charger. Using the Digital Multimeter on the 20 VDC range, measure the terminal voltage of the ATD-II Battery Module by placing the meter probes into the banana jacks on the ATD-II Battery Module. A reading of 12.6 to 14.4 VDC should be obtained. If the reading is below this value, it is likely that the ATD-II Battery Module is defective. See ILLUSTRATION 3-5.
4. Insert the Battery Module being tested into the ATD-II. Using the Endorectal Coil Simulator, cycle the tuning of the ATD-II as described in Section 3-4, TUNING TEST.
5. Leave the Endorectal Coil Simulator connected to the **PROBE INPUT** of the ATD-II for a period of 6.0 hours. Ensure that the ATD-II still displays the checkmark at the conclusion of the 6 hour period.
6. Disconnect the Endorectal Coil Simulator from the ATD-II. Insert the plug of the Endorectal Coil Simulator into the **PROBE INPUT** of the ATD-II and cycle the tuning of the ATD-II again as described in Section 3-4, TUNING TEST. If the display indicates **LOBAT**, the capacity of the Battery Module has decreased.
7. Remove the Battery Module from the ATD-II. Using the Digital Multimeter on the 20 VDC range, measure the terminal voltage of the ATD-II Battery Module. See ILLUSTRATION 3-5. A reading of 10.8 to 12.6 VDC should be obtained. If the reading is below this value, the capacity of the Battery Module has decreased. A replacement ATD-II Battery Module is needed. If the battery voltage is below 10.5 VDC, the ATD-II will display the **LOBAT** warning message.

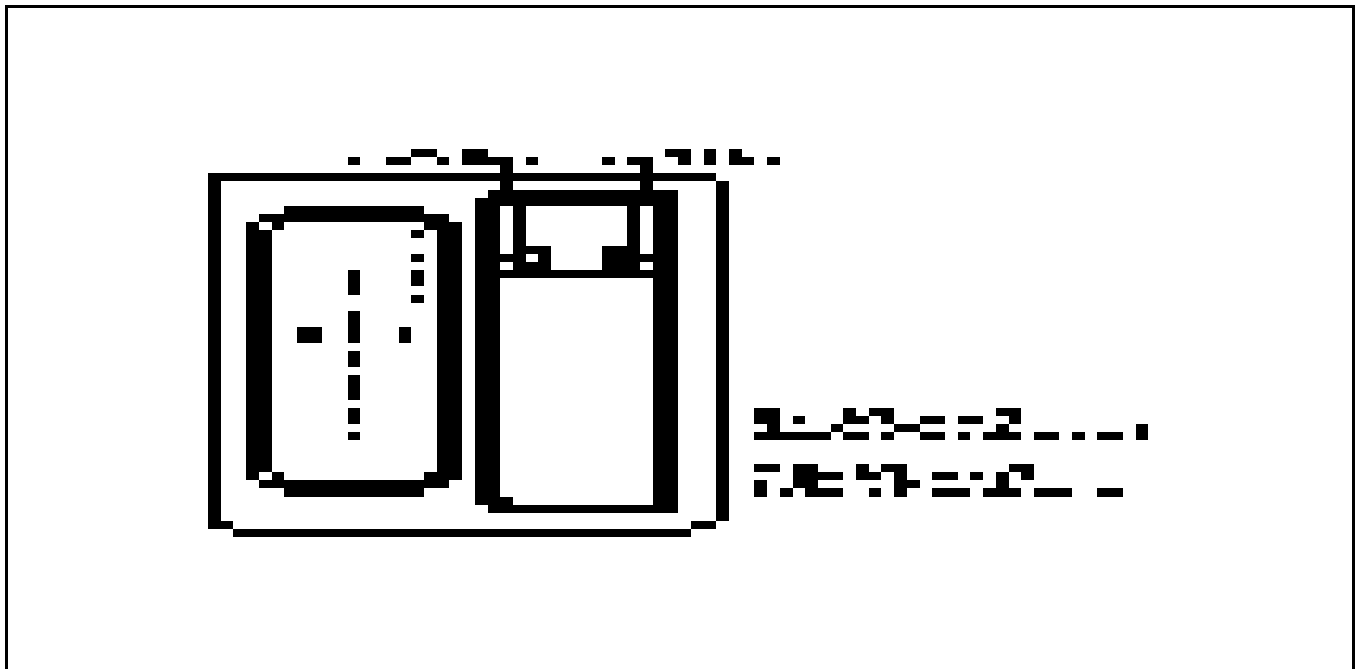


BATTERY MODULE
ILLUSTRATION 3-5

3-8 BATTERY CHARGER TEST**WARNING**

Do not bring the Battery Charger into the scan room. The Battery Charger contains many ferrous parts, possibly causing it to become a projectile in a strong magnetic field.

1. Connect the ATD-II Battery Charger to an appropriate AC power source.
2. Using the Digital Multimeter on the 20 VDC range, measure the terminal voltage of the ATD-II Battery Charger by placing the meter probes onto the banana plugs on the ATD-II Battery Charger. See ILLUSTRATION 3-6. A reading of 13.8 to 15.0 VDC should be obtained. If the reading is above or below this value, the ATD-II Battery Charger is defective.
3. Using the Digital Multimeter on the 2.0 ADC range, measure the current output of the ATD-II Battery Charger by placing the meter probes onto the banana plugs on the ATD-II Battery Charger. A reading of 0.350 to 0.700 ADC should be obtained. If the reading is above or below this value, the ATD-II Battery Charger is defective.



BATTERY CHARGER
ILLUSTRATION 3-6

SECTION 4 - REPLACEMENT/MAINTENANCE

4-1 ATD-II BATTERY MODULE REPLACEMENT

1. Open the battery door on the rear panel of the ATD-II, pull the old Battery Module out, and insert the new Battery Module. Replace and secure the battery door.

4-2 EXTERNAL COAXIAL CABLE REPLACEMENT

1. Remove the top cover of the ATD-II by removing the two slotted screws on the bottom surface of the device housing. The front panel will remain with the bottom cover. Retain the screws for reassembly.
2. Remove the ATD-II Front Panel and display PC board as a unit by pulling the assembly straight up. There are two 10 pin connectors at the bottom edge of the Display PC board which mate with two 10 pin plugs on the Main PC board. The connectors are not locking types; a vertical pull will separate them.
3. Loosen the strain relief body from the strain relief base. Slide the strain relief body down the coaxial cable a few inches.
4. Disconnect the cable by unscrewing the cable SMA connector from the SMA connector mounted to the Display PC board as shown in ILLUSTRATION 4-1.
5. Loosen and remove the strain relief base from the hole in the front panel. The base is held to the Front Panel by a nylon nut on the inside surface of the front panel of the device.
6. Remove the cable by passing the SMA connector out through the strain relief hole in the front panel. Discard the old cable.
7. Insert the new cable - Part Number 2100937-8, through the strain relief hole in the front panel. Place the nut from the strain relief over the SMA fitting. Route the cable to the SMA fitting on the Display PC board in the same fashion as the old cable.
8. Connect the cable by screwing the cable SMA connector into the SMA connector mounted on the Display PC board. Tighten the connector firmly. Do not use excessive force that could damage the connectors or the Display PC board.
9. Install the new strain relief base into the front panel. The base is fastened by a nylon nut inside the front panel. Use caution not to damage the finish of the front panel while installing and tightening the strain relief base.
10. Tighten the strain relief base. The convoluted tubing sheath should cover the cable completely. Loosen the strain relief body and adjust as needed.
11. Repeat the tests in Sections 3-2 and 3-3 to verify proper ATD-II operation.

4-2 EXTERNAL COAXIAL CABLE REPLACEMENT - Continued



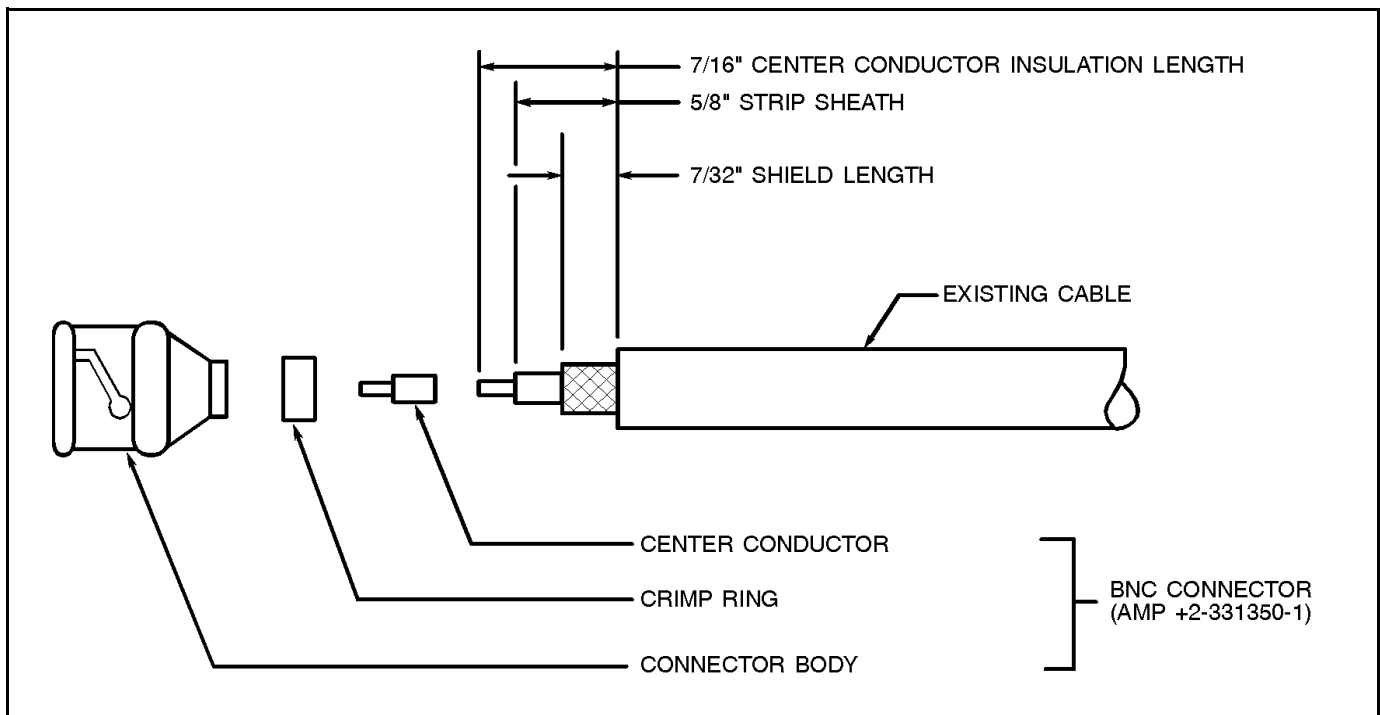
COAXIAL CABLE REPLACEMENT
ILLUSTRATION 4-1

4-3 BNC CONNECTOR REPLACEMENT

1. Prior to replacing the BNC fitting without completely replacing the coaxial cable, verify that the resulting cable length will be sufficient to permit convenient operation of the ATD-II and Coil. If the cable will be shortened more than 3" from original length by a repair, it is recommended that the entire cable be replaced in the event of a BNC fitting or fitting-end cable failure.
2. Remove the old shrink-tubing from the BNC connector and convoluted tubing. Place 1.5 inch piece of shrink tubing over the convoluted tubing. Pull back convoluted tubing. Replace the BNC connector using ILLUSTRATION 4-2 as a guide. Use a new BNC coaxial fitting, part number 46-271494P1, and crimp tool part number 46-255841 with insert part number 46-255841P100 to install the replacement connector on the cable. Push unshrunk tubing up to the base of the BNC connector, push the convoluted tubing under the unshrunk tubing. Apply heat and shrink tubing, ensuring that both the BNC connector and convoluted tubing are captured.
3. Repeat the tests in Sections 3-2 and 3-3 to verify proper ATD-II and Coil operation.

4-4 DIODE REPLACEMENT

1. ATD-II calibration and operation may be affected by diode replacement. Field replacement of the decoupling or RF protection diodes is not recommended.



BNC CONNECTOR REPLACEMENT
ILLUSTRATION 4-2

4-5 REPLACING THE MECHANICAL HARDWARE

1. Replace the entire ATD-II. Refer to Section 5, RENEWAL PARTS, for ATD-II part number.

4-6 CHECKING THE EXTERNAL CABLE

1. Check the external cable for cracks or breaks once each week. Replace the external cable if any damage or wear is found. Refer to Section 4-2, EXTERNAL COAXIAL CABLE REPLACEMENT, for replacement procedure.

4-7 CLEANING THE ATD-II

1. Clean the ATD-II and external cable with a mild dishwashing liquid and water solution. Wet a soft cloth with the solution and proceed to clean.

SECTION 5 - RENEWAL PARTS

5-1 SERVICE KITS

1. The Endorectal Coil Simulator part number is:

ITEM	DESCRIPTION	QUANTITY
1	Endorectal Soil Simulator - 2100937-31	

2. The ATD-II part number is:

ITEM	DESCRIPTION	QUANTITY
1	ATD-II with Cable-45-328293P1	1

3. The Service Kit for cable assembly replacement part number is:

ITEM	DESCRIPTION	QUANTITY
1	Cable Kit-2100937-8	1

SNR QA DATA TABLE

Use the space provided below to record the calculated signal to noise ratio—SNR, data obtained from Section 4, *SNR IMAGE ANALYSIS*. After recording the SNR data obtained during the initial coil installation, record subsequent SNR data in column 2, enter the date in column 1 as a periodic QA check. If the ratio of any of the coils found is not greater than 85%, then there is a problem in the coil or the MR system.

Original SNR data obtained at initial coil installation:

Body Coil SNR Value: _____

ATD-II SNR value: _____

Date: _____

SNR DATA QA (QUALITY ASSURANCE) CHECK:

1	2	3	
Date	ATD-II QA SNR Value:	Is new value divided by original value >85%?	
_____	_____	_____	Y/N
_____	_____	_____	Y/N
_____	_____	_____	Y/N
_____	_____	_____	Y/N
_____	_____	_____	Y/N
_____	_____	_____	Y/N
_____	_____	_____	Y/N
_____	_____	_____	Y/N
_____	_____	_____	Y/N
_____	_____	_____	Y/N
_____	_____	_____	Y/N

Make additional copies of this document as needed.

DEFECTIVE SURFACE COIL RETURN FORM

NOTE

For proper assessment of defective returned coils, both sides of this form should be completely filled out and accompany all returned coils. Include films or prints of any image quality complaints with a description of the scan prescription used.

DATE: _____

SITE NAME: _____

SITE ADDRESS: _____

SERVICE ENGINEER: _____

COIL SERIAL NUMBER: _____

DATE COIL INSTALLED: _____

DESCRIPTION OF COIL PROBLEM: _____

(Please be descriptive; "Broken" is not enough) _____

Note

An alternate proprietary procedure is available to GE use and to customers with a valid Advanced Service Package Limited License. Refer to "TLT PROCEDURE" located in Direction 15491, *Signa Advantage 1.5T, 1.0T & 0.5T System Troubleshooting Guide*. Use the table to record the TLT results of the defective coil.

TABLE DR-1
TLT DATA FOR DEFECTIVE SURFACE COIL

SITE:		NAME:	DATE:
TLT FILE NUMBER:			
SNR:	NOISE		
	SNR MEAN		
	SNR SIGNAL		
	SNR AREA		
TRMAP:	89-91	%	%
	85-95	%	%
	65-115	%	%
	FLIP MEAN		
	FLIP SDV		
	RCV MEAN		
	RCV SDV		
	DIFF MEAN		
	DIFF SDV		



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